Baseline Reproduction – Technical Memo

Team: SixtyFive.AI (Group Number 65)

Date: October 19th, 2026

Subject: EuroSAT Geographic Domain-Shift Classification

1. Repository Submission

Repository: https://github.com/Rayan-RK/CSCI3052U-Project-TeamSixtyFive/tree/main
Release tag:

2. Reproducible Environment & Repository Setup

The implementation is on the main branch. Install dependencies with python -m pip install -r requirements.txt (Python 3.11 or compatible). The README documents data downloads, directory layout and execution commands. Baselines use RANDOM_SEED = 42 for the stratified split and stochastic model components. Exact dependency versions and an independently verified fresh-environment rerun are not yet documented; a tagged release remains pending.

3. Working Data Pipeline

- **Ingestion and verification:** EuroSAT v2 provides 27,000 paired RGB/multispectral samples across 10 land-cover classes. M2 EDA checks archived data integrity, image shapes and geographic metadata.
- **Geographic assignment:** src/build_geographic_manifest.py maps GeoTIFF coordinates to country polygons. Western/Central Europe is development; Southern/Eastern Europe is candidate OOD. Points within 1 km of country boundaries or without unambiguous matches are flagged/excluded from these two candidate partitions.
- **Splitting and preprocessing:** Development samples are split 80/20 with class stratification (seed 42). RGB images are resized to 16 × 16, flattened into 768 values and divided by 255. PCA is fitted only on the training subset inside a scikit-learn pipeline.

4. Baseline Methodology

- **Naive benchmark:** DummyClassifier(strategy="most_frequent") always predicts the most common development training class; no image features are used.
- **Course-level baseline:** PCA(n_components=100) followed by RandomForestClassifier(n_estimators=200, class_weight="balanced", random_state=42). Both are evaluated on development validation and held-out candidate OOD samples. Macro-F1 is primary; accuracy is supplementary.

5. Baseline Table

Measured results from outputs/combined_baseline_results.csv:

Model	Partition (n)	Accuracy	Macro-F1
Majority class	Development validation (3,516)	0.1359	0.0239
Majority class	OOD test (6,231)	0.0913	0.0167
PCA + Random Forest	Development validation (3,516)	0.7085	0.6910
PCA + Random Forest	OOD test (6,231)	0.4486	0.4365

6. Discrepancy Analysis

The RGB PCA + Random Forest Macro-F1 falls from 0.6910 on development validation to 0.4365 on OOD data (absolute gap 0.2546); accuracy falls from 0.7085 to 0.4486. The majority-class benchmark also declines (Macro-F1 0.0239 to 0.0167). Per-class F1 drops are largest for Pasture (0.5828) and Herbaceous Vegetation (0.4588), but small for Forest (0.0524) and Sea/Lake (0.0216). These are observed within-project discrepancies, not a verified reproduction gap against a published reference score. Different class mixes (e.g., Pasture: 374 validation vs 54 OOD samples), preliminary geographic assignments, and random rather than geographically grouped development validation limit causal interpretation.

7. Start / Stop / Continue Plan

- **Start:** Audit spatial overlap and finalize geographically grouped validation/frozen manifests; pin package versions and verify a clean rerun before stronger OOD claims.
- **Stop:** Treating a random stratified validation score or the provisional country-boundary screen as proof of leakage-free geographic generalization.
- **Continue:** Maintaining modular scripts, fixed seeds, train-only PCA fitting, Macro-F1 comparisons and per-class error reporting while extending to later models.